Loops with multiple range() parameters

Earlier, you learned about the range function and how it generates a sequence of numbers starting with zero. However, your work as a data professional, won't always need to start with zero. Instead, you might want to use a step parameter in the range function. This makes it possible to specify which elements of a variable or dataset we want to start and end with. In this video, you'll learn how to use the three parameters of the range function: start value, stop value, and step value. Here's a for loop that calculates the factorial of nine. Let's explore how it works.

We begin by assigning a variable called "product" with the number one. Next we have a range function that starts at one and stops at 10. Since the stop value is not included in the calculation, this means the code multiplies the variable by every whole number in the sequence beginning with one and ending with nine. So, for n in range one to 10, we multiply our "product" variable by one and reassign the result back to itself. Then, we multiply the "product" variable by two and reassign the result back to itself, then by three... all the way up to nine. This produces the factorial of nine. The factorial of nine equals 362,880.

Note that we started with one and not with zero. If we multiplied by zero, the product would be zero. The RANGE function also lets us specify a third parameter to change the size of each step in our range. In other words, instead of going one by one, you can have a larger difference between each of the values in your range. Let's explore an example. First, define a function that converts a temperature value from Fahrenheit to Celsius. Use the standard conversion formula.

The temperature in Fahrenheit that needs to be converted is identified by x, 32 is subtracted from x, times 5, divided by 9. Next we have a for loop that will print a table of temperature conversions every 10 degrees from 0 to 100 degrees Fahrenheit. Notice that the for loop starts at zero and goes up to 100 in steps of 10. Remember that the range excludes the final value in a sequence, so to include 100 in our sequence, we put the end value as 101. The body of the for loop prints the value in Fahrenheit and the value in Celsius to create a conversion table. For every 10 degrees Fahrenheit, the code prints the corresponding value in Celsius on each line. So, now you know more about setting parameters for the range function when you're working with for loops.

As a quick reminder, use for loops when there's a sequence of elements that you want to iterate over. For example, to loop over a variable, such as a record in a dataset, it's always better to use for loops. This also improves the readability of your code. Use while loops when you want to repeat an action until a boolean condition changes, without having to write the same code repeatedly. Remember, booleans are a data type that represents one of two possible states: usually True or False. And, if whatever you're trying to do can be done with either a for loop or a while loop, just use whatever you prefer. I happen to love them both and am really glad I have both tools in my Python toolbox!

Parameters of the range function

- Start value
- Stop value
- Step value

Use **for loops** when there's a sequence of elements that you want to iterate over.

Use **while loops** when you want to repeat an action until a boolean condition changes.

Booleans are a data type that represents one of two possible states: **True** or **False**.

Question

Python's `range()` function includes which of the following parameters? Select all that apply.

- [ ] Loop value
- [x] Stop value

Correct
Python's `range()` function includes the following parameters: start value, stop value, and step value. The `range()` function returns a sequence of numbers starting from zero; then increments by one, by default; then stops before the given number.

- [x] Step value

Correct
Python's `range()` function includes the following parameters: start value, stop value, and step value. The `range()` function returns a sequence of numbers starting from zero; then increments by one, by default; then stops before the given number.

- [x] Start value

Correct
Python's `range()` function includes the following parameters: start value, stop value, and step value. The `range()` function returns a sequence of numbers starting from zero; then increments by one, by default; then stops before the given number.

Continue

Skip